

```
# immutable  
# data type==< collect for data
```

```
x =10,20  
print(x)
```

```
(10, 20)
```

```
x =10,20  
print(type(x))
```

```
<class 'tuple'>
```

```
x =(10,20,"divya")  
print(type(x))
```

```
x =(10,20,"divya")  
print(x)
```

```
<class 'tuple'>  
(10, 10, 20, 'divya')
```

```
# we can,t do insert,delete,update  
x=10,10,10,20,"divya","rahul"  
print(x.count(10))  
print(x.index("divya"))
```

```
3  
4
```

```
# list vs tuple  
# [],()
```

```
# dictionary : datatype,use to store data but

# without index position
# data is store => key,Values
# ,key is unique
# value can be unique/duplicate
# dictionary are Mutable
# no index position
```

```
mydictionary={101:"divya","salary":3452,"email":"tushar0"}
print(mydictionary["salary"])
```

3452

```
mydictionary={101:"divya","salary":3452,"email":"tushar0"}
print(mydictionary["salary"])
mydictionary["email"]="bansaldivya@186gmail.com"
print(mydictionary['email'])      # update
mydictionary["email"]="jaipur"
print(mydictionary)
```

3452

bansaldivya@186gmail.com

{101: 'divya', 'salary': 3452, 'email': 'jaipur'}

```
mydictionary.keys()
mydictionary.values()
```

dict_values(['divya', 3452, 'jaipur'])

```
x=mydictionary.pop("salary")
print(mydictionary,x)
```

{'email': 'jaipur'} 3452

```
mydictionary={10:"divya",34:"rahul"}  
for i in mydictionary:  
    print(i,mydictionary[i])
```

```
10 divya  
34 rahul
```

```
mydictionary={10:"divya",34:"rahul"}  
for i in mydictionary.keys():  
    print(i)
```

```
10  
34
```

```
mydictionary={10:"divya",34:"rahul"}  
for i in mydictionary.values():  
    print(i)
```

```
divya  
rahul
```

```
mydictionary={10:"divya","rahul":45}  
mydictionary["rahul"]=mydictionary["rahul"]+5  
print(mydictionary)
```

```
{10: 'divya', 'rahul': 50}
```

```
s="abc"  
for i in "abc":  
    mydictionary[i]=1  
    print(mydictionary)
```

```
{10: 'divya', 'rahul': 50, 'a': 1}  
{10: 'divya', 'rahul': 50, 'a': 1, 'b': 1}  
{10: 'divya', 'rahul': 50, 'a': 1, 'b': 1, 'c': 1}
```

